

stressEngine.ts

frontend/utils | TypeScript

```

/**
 * stressEngine.ts
 *
 * Core cognitive load computation engine.
 *
 * Implements the Weighted Fusion Model that combines three independent
 * biosignal channels into a single Cognitive Score, and the State Machine
 * that maps scores to OS states with hysteresis.
 *
 * Fusion formula:
 * Score = (0.5 × vocalJitter) + (0.3 × rPPG Variance) + (0.2 × semanticIntent)
 *
 * Weight rationale:
 * - Audio (0.5): Vocal jitter is the most direct real-time cortisol proxy
 * available without clinical hardware. Highest reliability.
 * - Visual (0.3): rPPG is scientifically validated but consumer webcam SNR
 * varies significantly with lighting. Moderate weight.
 * - Semantic (0.2): Intent classification is discrete and event-driven
 * (requires active speech). Lowest weight; provides contextual tiebreaking.
 */

// ■■■ Types ■■■
export type CognitiveState = 'CALM' | 'LOAD' | 'CRITICAL';

export interface FusionInputs {
  /** Vocal jitter score, normalized 0-100 (from useVocalJitter hook) */
  vocalJitter: number;
  /** rPPG variance score, normalized 0-100 (from useRPPG hook) */
  rppgVariance: number;
  /** Semantic intent score, normalized 0-100 (from hinglishParser) */
  semanticIntent: number;
}

export interface CognitiveAssessment {
  score: number; // Computed fusion score (0-100)
  state: CognitiveState; // Current OS state
  transitioned: boolean; // Whether state changed this update
  prevState: CognitiveState; // Previous state (for transition animations)
}

// ■■■ Fusion Weights ■■■

/** Standard weights – all three channels operating normally */
const WEIGHTS = {
  audio: 0.5,
  visual: 0.3,
  semantic: 0.2,
} as const;

/**
 * Adaptive weights – activated when rPPG signal confidence < 60%.
 *
 * When lighting degrades or the user leaves the camera frame, rPPG SNR drops.
 * Rather than continuing to weight a noisy visual channel at 0.3, we:
 * - Drop visual weight from 0.3 → 0.1 (still contributes, but minimally)
 * - Boost semantic from 0.2 → 0.4 (Hinglish NLP gains primary support role)
 * - Keep audio at 0.5 (jitter is unaffected by lighting)
 *
 * Reference: documented in docs/architecture.md §2.2 and docs/signal-processing.md §Confidence
 */
const ADAPTIVE_WEIGHTS = {
  audio: 0.5,
  visual: 0.1,
  semantic: 0.4,
}

```


[illegible]